

Water Segmentation using Conventional Method

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Initial Approach

Watershed

First, the input image is converted to grayscale, as many image processing operations, such as thresholding, are designed for single-channel images. Second, Otsu's Method automatically determines an optimal threshold value, segmenting the image into a binary mask. In this process, inverse thresholding assigns pixels below the threshold a value of 255 (white) to represent the foreground, while those above the threshold are assigned a value of 0 (black) to represent the background.

Third, morphological refinement is applied using a 5x5 elliptical kernel with a closing operation to eliminate small holes and smooth the boundaries of the foreground. Fourth, a distance transform calculates the distance of each pixel in the binary mask to the nearest zero pixel, assigning higher values to pixels that are closer to the center of the foreground regions. Thresholding retains pixels that exceed 40% of the maximum distance as the "sure foreground," which is then converted to an 8-bit format for further processing. Fifth, dilation expands the foreground regions to define the "sure background," and subtracting the foreground from the background identifies the overlapping areas as the "unknown" region.

Sixth, the connected components algorithm assigns unique labels to each region in the "sure foreground," while background pixels are initially labeled as 0. This label is then incremented by 1 to shift the background label to 1, and pixels in the unknown region are reassigned to 0. Seventh, the watershed algorithm treats the image as a topographical surface and floods it from these labeled markers. Eighth, boundaries where two regions meet are labeled as -1. The binary segmentation result is generated by setting pixels with labels greater than 1 to white (255) for the segmented regions, while pixels marked as -1 (boundaries) are labeled as black (0). Finally, the processed binary image, in which segmented regions are white and boundaries are black, is returned as the output.

However, upon examining the output of the watershed algorithm, I find that while the algorithm can reasonably predict the edges of the objects in the input image, it struggles to accurately differentiate between water and non-water areas. Consequently, we plan to develop a statistical model, known as the Maximum A Posteriori (MAP) estimation, to address this issue more effectively.

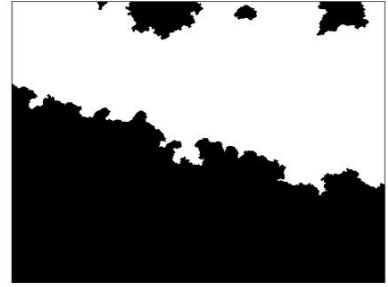
image



actual mask



predicted mask



Intersection over Union (IoU) : **0.62**

image



actual mask



predicted mask



Intersection over Union (IoU) : **0.40**

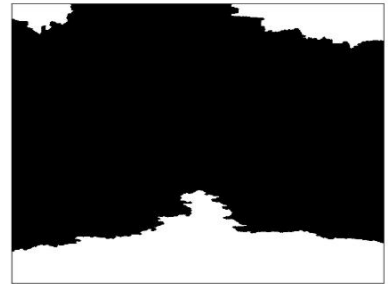
image



actual mask



predicted mask

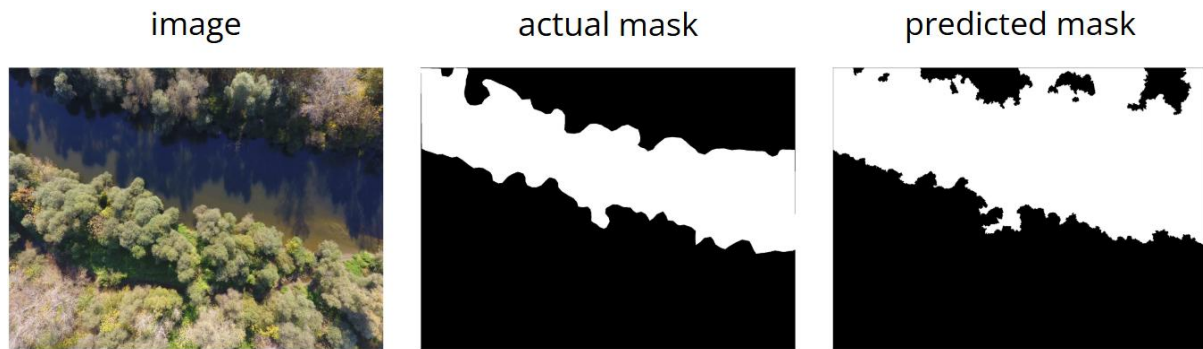


Intersection over Union (IoU) : **0.40**

Qualitative/Quantitative Analysis:

Upon examining the output of the watershed algorithm, we find that while it can reasonably predict the edges of objects in the input image, it struggles to accurately differentiate between water and non-water areas, resulting in suboptimal IoU values. Therefore, we plan to apply a statistical model known as MAP estimation to address this issue more effectively.

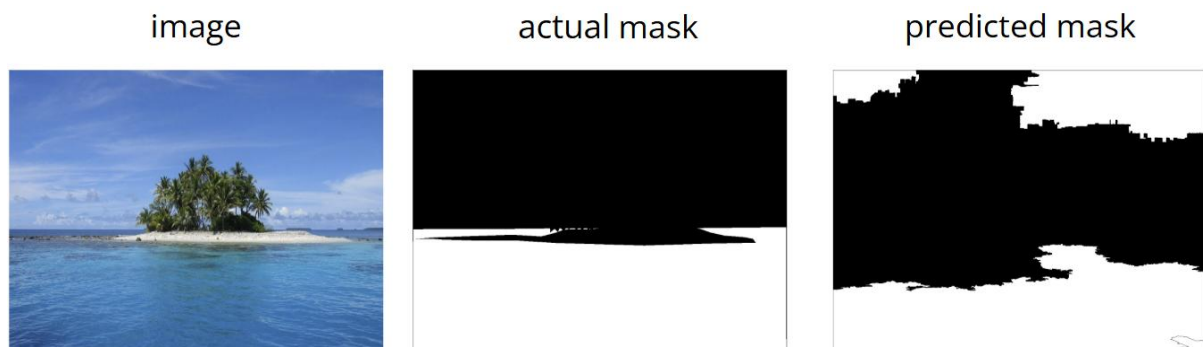
Watershed with MAP Model



Intersection over Union (IoU) : **0.64**



Intersection over Union (IoU) : **0.45**



Intersection over Union (IoU) : **0.52**

Qualitative/Quantitative Analysis:

By analyzing the output of the watershed algorithm in conjunction with the MAP model, it is evident that the distinction between water and non-water areas has become more precise; however, the average IoU values have not shown significant improvement. Besides, the discrepancies between the actual and predicted masks indicate that there is still an inaccurate assessment of brightness changes on the water's surface and a limited capacity to differentiate between similar colors, such as water and sky.

Morphological Method

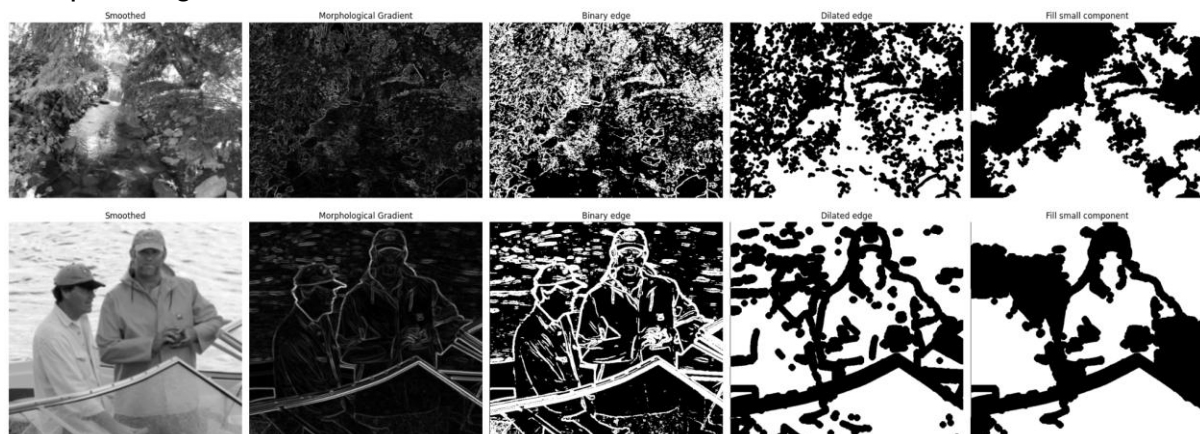
Initially, I assumed that water regions would typically appear smooth, while the surrounding areas would exhibit greater complexity. Based on this assumption, I approached the problem by distinguishing between smooth and non-smooth regions.

First, I converted the image to grayscale to simplify processing and focus on intensity values. To detect image boundaries, I applied Gaussian smoothing followed by a morphological gradient. The morphological gradient, obtained by subtracting the eroded image from the dilated image, highlights regions with rapid intensity changes, typically corresponding to boundaries. I then transformed the grayscale image into a binary image, using the mean intensity of the morphological gradient as the threshold.

Next, I performed a binary opening operation to remove small edge artifacts, followed by dilation to close gaps between adjacent boundaries. Finally, I identified all connected components in the image. Regions smaller than a predefined threshold were classified as the same region as the black ground.

This algorithm performed well under the initial assumption that water regions are smooth and the surroundings are more complex. However, it became evident that this assumption was overly simplistic. For example, water regions may contain waves, and smooth areas such as the sky can lead to incorrect predictions, significantly impacting the algorithm's accuracy.

Sample Image



MAP model

To address the problem, I decided to develop a statistical model to classify enclosed regions within the image. By leveraging the likelihood function and incorporating prior information derived from the provided training data and mask, I constructed a Maximum A Posteriori (MAP) model. This approach allowed me to estimate the most probable classification for each region by balancing the likelihood of the observed data with the prior probabilities.

Feature Extraction

For feature extraction, I selected features primarily based on color and texture, considering that the boundaries, area, and compactness of the water regions can exhibit significant variance. To accurately capture these characteristics, I implemented a set of methods to extract meaningful features from the image and its corresponding mask.

Color Features:

To analyze the color properties of the water regions, I used the provided mask to isolate the relevant parts of the image. The image was converted to the HSV color space, which is better suited for distinguishing color characteristics under varying lighting conditions. I computed the mean and standard deviation for each channel in both the RGB and HSV color spaces. These features capture the average color distribution and variability within the masked region, providing a robust representation of the water's color properties.

Texture Features:

Texture analysis was performed on the grayscale version of the image to capture structural patterns within the water regions. I utilized the following techniques:

Local Binary Patterns (LBP): LBP was applied to identify local texture patterns. The resulting histogram, normalized for density, summarizes the frequency of different patterns within the masked region.

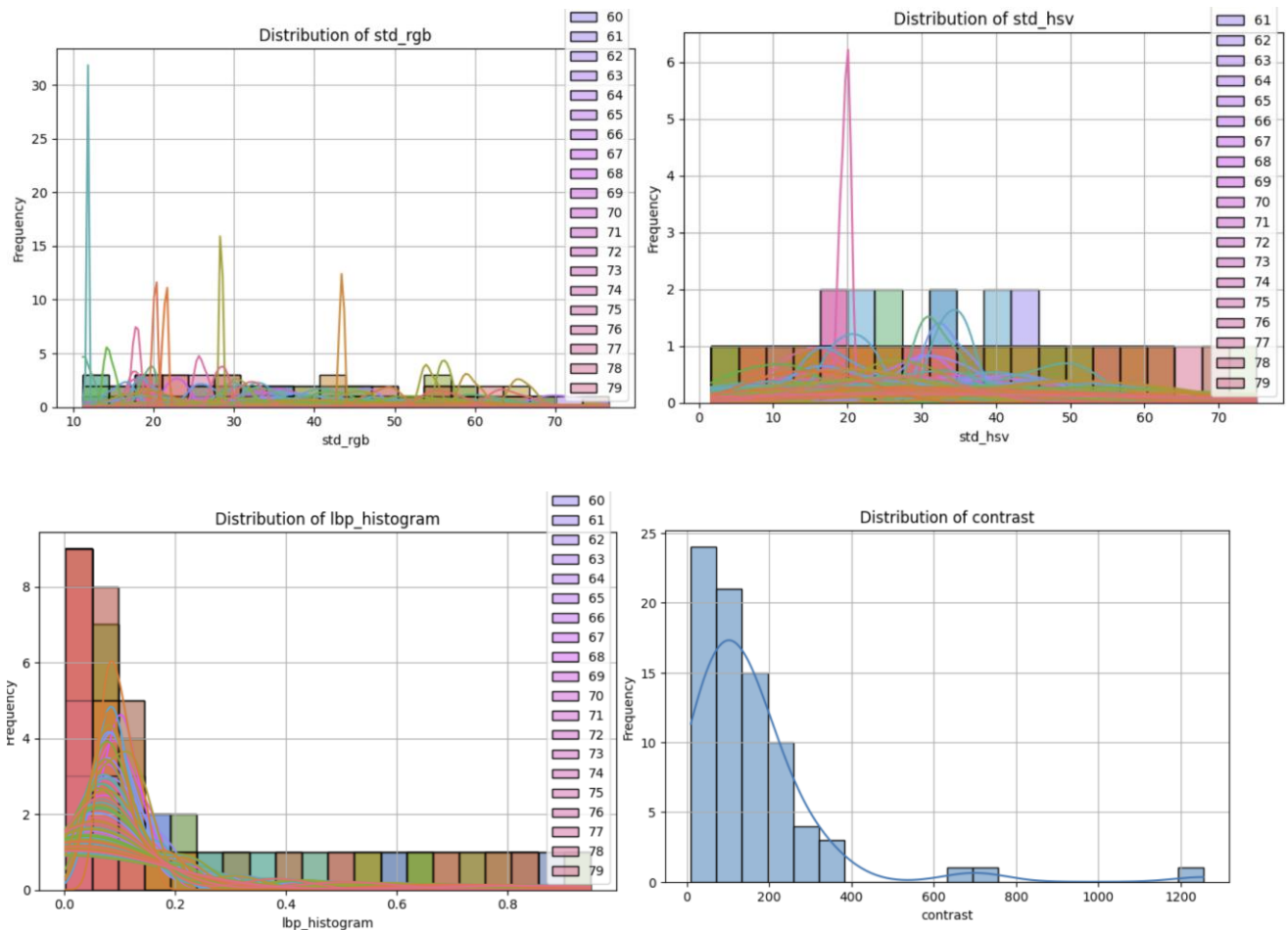
Haralick Features: Using the Gray-Level Co-occurrence Matrix (GLCM), I extracted four statistical texture measures: contrast, energy, homogeneity, and correlation. These features provide insights into the texture's spatial relationships and uniformity.

Entropy: Shannon entropy was calculated to measure the complexity of the texture in the masked region.

Feature Combination:

The extracted color and texture features were combined into a single feature set to serve as input for the MAP model. This combination ensures that both the visual appearance and structural properties of the water regions are captured effectively.

Sample Feature Distribution



Model Training

To train the Maximum A Posteriori (MAP) model, I utilized a statistical approach that relies on prior probabilities and likelihood estimation. The training process is summarized as follows:

Class Separation:

The input dataset was divided into two classes: "water" and "non-water." Features corresponding to each class were grouped to enable class-specific analysis.

Prior Estimation:

The prior probabilities for each class were calculated based on their frequency in the dataset. For instance, if the dataset contained more water regions than non-water regions, the prior probability for the "water" class would be higher.

Likelihood Estimation:

Assuming a Gaussian distribution for simplicity, the likelihoods of each feature were estimated for both classes. For each feature, the mean and standard deviation were computed across all instances of the respective class. These parameters define the Gaussian distribution, which is used to calculate the likelihood of a given feature value belonging to a particular class.

Prediction with the MAP Model:

During inference, the model computes the posterior probability for each class using Bayes' theorem. The class with the highest posterior probability is chosen as the prediction. For numerical stability, log probabilities are used in the computation.

Result and Comparison

After training, we evaluated the accuracy of the MAP model, achieving 0.66 on the training image and 0.78 on the testing image. While these results indicate the model's ability to distinguish between water and non-water regions to some extent, the accuracy is relatively low. Below are some possible reasons for this performance:

1. Choice of Likelihood Function:

The Gaussian distribution was selected as the likelihood function for feature modeling. While convenient, this assumption may not accurately represent the distributions of the selected features. Real-world data often exhibit more complex distributions that cannot be fully captured by a simple Gaussian model.

2. Feature Selection Process:

Unlike deep learning models, which automatically learn features from the training data, the MAP model relies on manually selected features. The effectiveness of the MAP model is heavily dependent on the quality and relevance of these features. In this case, the selected features may not fully capture the variability and discriminative properties required for accurate classification.

Despite these limitations, the MAP model offers significant advantages in terms of training time and data requirements. Unlike deep learning models, which require large datasets and computationally expensive training processes, the MAP model only needs to estimate the mean and variance for the likelihood function. This simplicity allows the MAP model to be trained quickly, even with a limited amount of data.

Final Algorithm

Data Process Procedure

1. Train MAP model using training data and provided mask.
2. Segment the image based on the edges.
 - a. Convert the png image into a grayscale array.
 - b. Apply Gaussian Smoothing to reduce noise and detail.
 - c. Calculate the morphological gradient to highlight edges.
 - d. Determine the threshold for the image based on its mean and standard deviation.
 - e. Connect edgels using multiple thresholds obtained in step d.
 - f. Apply a binary closing operation to remove small edgel artifacts and smooth the edges.
 - g. Identify connected elements; if an element is too small, classify it as part of the background.)
3. Classify each connected element using the MAP model.
4. Fail prevention
 - a. If the MAP model classifies all elements as non-water, output the boundary mask instead.

Main Improvements

Edge Connection Using Thresholds:

Instead of relying on the mean to transform the grayscale morphological gradient image and using dilation to merge edges as in the original MAP method, I connected the edges by comparing binary image transformations generated using multiple thresholds. These thresholds were calculated based on the gradient's mean and standard deviation.

Eliminating Assumptions:

Unlike the original approach, no assumptions were made about the characteristics of water and non-water regions.

Segmentation and Classification:

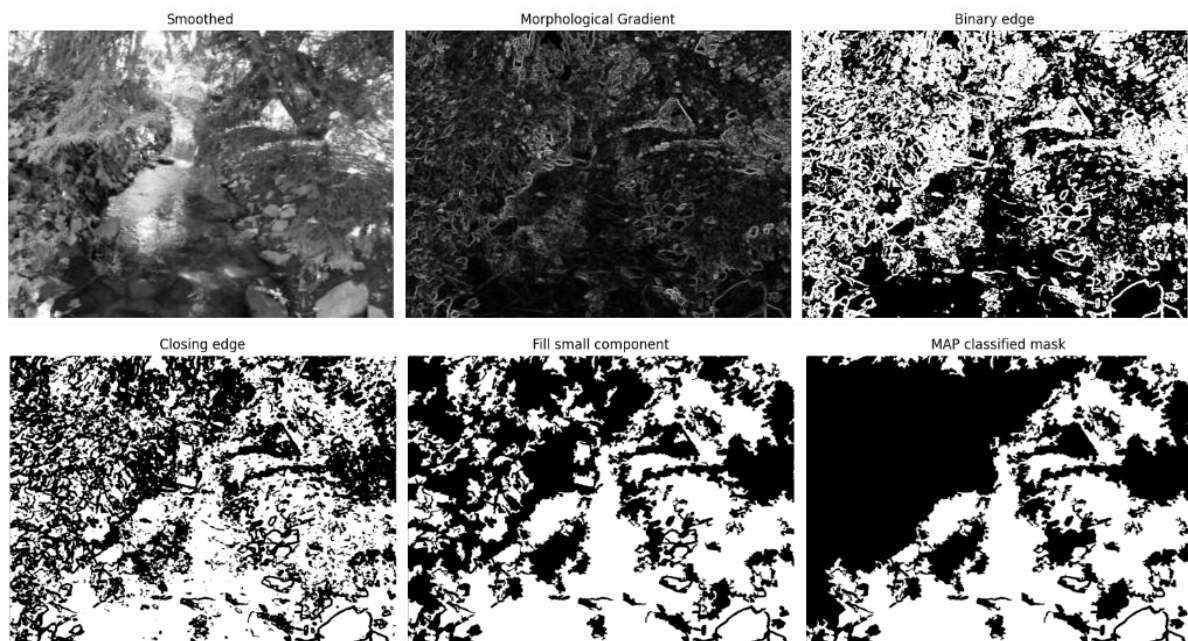
Morphological operations were applied to extract edges and divide the image into segments. The MAP model was then used to classify each segment.

Fail Prevention:

During testing, we observed that the model occasionally failed, producing an entirely black mask. In such cases, the initial mask generated by the morphological method was used as the output mask to ensure functionality.

Experimental Results

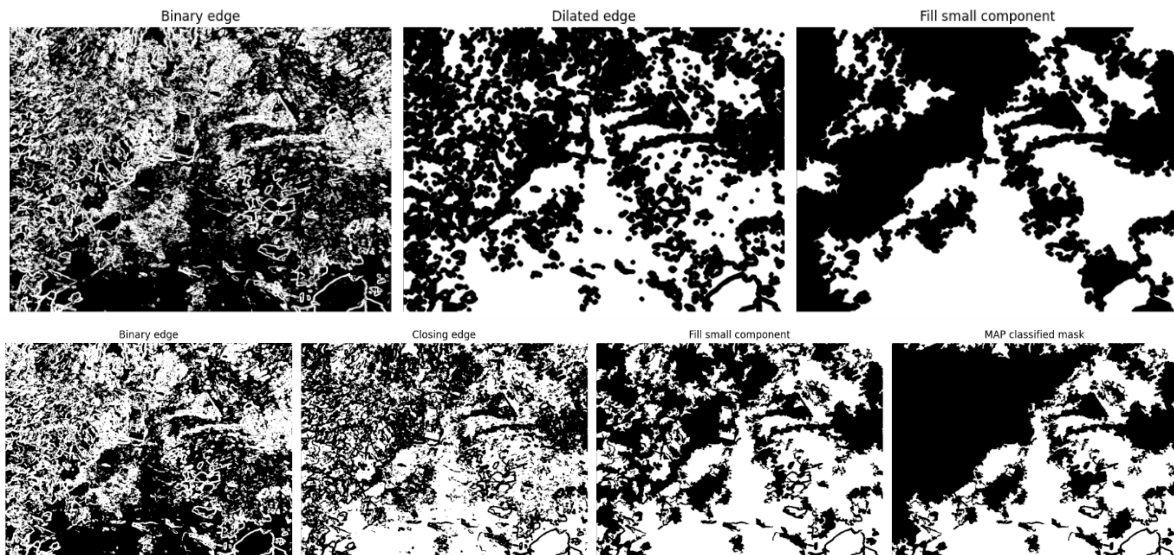
Sample Image 1



Sample Image 2



Result Comparison (pure morphological / final model)



IoU: 0.366951/ 0.3856



IoU: 0.309660 / 0.365411

Possible reasons that cause poor accuracy

The poor accuracy of the algorithm can be attributed to several factors. First, the assumption of Gaussian distribution for the likelihood function may not adequately represent the true distributions of the selected features, leading to suboptimal classification. Second, the manually selected features, while capturing some aspects of color and texture, may not fully encompass the variability and discriminative properties necessary for accurate segmentation. Additionally, the algorithm struggles to differentiate between visually similar regions, such as water and sky, especially

under varying lighting conditions or when brightness changes occur on the water's surface. The reliance on thresholds derived from global statistics, rather than adaptive or localized criteria, further limits its ability to handle diverse scenarios. Lastly, the simplicity of the MAP model, while efficient, lacks the capacity to learn complex patterns automatically, which deep learning-based approaches might achieve more effectively.

Conclusion

In this project, we explored multiple methods for water segmentation, including the watershed algorithm, morphological operations, and the Maximum A Posteriori (MAP) model. While the watershed algorithm provided a foundation for edge detection, its limitations in distinguishing water from non-water regions highlighted the need for a more robust approach. The morphological method introduced the concept of smoothness to segment water regions, but it was overly simplistic and prone to errors in complex scenarios.

The MAP model, incorporating statistical features like color and texture, improved classification accuracy by leveraging prior probabilities and likelihood estimation. Despite achieving higher Intersection over Union (IoU) scores compared to earlier methods, the model's reliance on Gaussian assumptions and manually selected features constrained its performance. Nonetheless, the MAP model demonstrated significant advantages in terms of efficiency and data requirements, offering a practical solution for segmentation tasks with limited datasets.

Our final approach integrated edge detection with the MAP model, addressing some of the observed shortcomings. However, challenges such as distinguishing between visually similar regions, like water and sky, and handling brightness variations remain areas for improvement.

Overall, this project provided valuable insights into the complexities of water segmentation and highlighted the trade-offs between accuracy, computational efficiency, and data dependency in algorithm design.